

Ex 1: Install Virtualbox/VMware/ Equivalent open source cloud Workstation with different flavours of Linux or Windows OS on top of windows 8 and above.

Aim:

To install VirtualBox, VMware, or an equivalent open-source virtualization tool on a Windows 8 or later system and set up virtual machines with different Linux or Windows OS flavors.

Requirements

- A computer with Windows 8 or above
- Virtualization software: Oracle VirtualBox / VMware Workstation / Equivalent open-source software
- ISO images of Linux (Ubuntu, CentOS, Fedora, etc.) or Windows OS
- Minimum 8GB RAM and 50GB free disk space

Concept:

Virtualization is a key concept in cloud computing, allowing multiple operating systems to run on a single physical machine. Tools like VirtualBox and VMware help create isolated environments known as virtual machines (VMs), which can be used for cloud-based development, testing, and deployment.

Steps to Install VirtualBox or VMware

Step 1: Download and Install Virtualization Software

1. Download VirtualBox from <https://www.virtualbox.org/> or VMware Workstation from <https://www.vmware.com/>.
2. Run the installer and follow on-screen instructions.
3. Restart the system if required after installation.

Step 2: Create a New Virtual Machine (VM)

1. Open VirtualBox or VMware Workstation.
2. Click **New** to create a new VM.
3. Assign a name and select the operating system type (Linux or Windows).
4. Allocate RAM (Recommended: 4GB for Linux, 8GB for Windows).
5. Create a virtual hard disk (Minimum 20GB recommended).

Step 3: Install OS on the Virtual Machine

1. Select the newly created VM and click **Start**.
2. Browse and select the ISO image of the OS to install.
3. Follow the OS installation steps.

4. After installation, install guest additions/tools for better performance.

Step 4: Configuring Network and Shared Folders

1. Configure **Bridged Adapter** for internet access.
2. Enable shared folders to share files between host and VM.

Step 5: Testing the Virtual Machine

1. Verify the installation by booting into the VM.
2. Check network connectivity and shared folder access.
3. Install essential tools and software required for cloud computing tasks.

Observations

- Note down the OS flavors installed.
- Record system resource usage for different VMs.
- Test system performance with multiple VMs running.

Result

Upon completing this experiment, students will have successfully installed and configured a virtual machine, laying the foundation for cloud-based environments.

Viva Questions:

1. What is virtualization, and why is it important for cloud computing?
2. What are the differences between VirtualBox and VMware?
3. How does a virtual machine differ from a physical machine?
4. What are the advantages of using different OS flavors in virtualization?
5. What role do guest additions or VMware tools play in virtualization?